

Target Regions and Geological Context

【35†embed_image】 Cornwall's mineral wealth arises from its complex Variscan geology. The 250 km-wide **Cornubian Batholith** of Permian age underlies much of Cornwall, intruding late Devonian/Carboniferous slates 【73†L674-L679】. This orogenic event caused folding and fracturing of the old sediments and emplacement of coarse granites, which in turn drove extensive metal-bearing hydrothermal veins (Cu–Sn–W–As) and superimposed kaolinisation of the granite roofs 【73†L674-L679】 【35†L670-L678】. Major granite bodies (e.g. St Austell, Bodmin Moor, Land's End, Carnmenellis) host *primary* mineralization (Sn, W, Li-mica, REE in micas) and *secondary* kaolin (china clay) deposits 【73†L674-L679】 【70†L975-L983】. Figure above shows a simplified geology map with key mineral commodities. Cornwall's minerals span base metals (Cu, Pb, Zn), tin-group (Sn, W, As, Sb, Bi), lithium in micas, and vast kaolin resources 【39†L21-L27】 【70†L975-L983】. Precious metals (Au, Ag) and other critical elements (Ni, Co) are present but less explored 【81†L100-L103】.

Historical Production and Resources

Cornwall's mining history dates to Bronze Age tin working 【60†L740-L748】. Production **peaked in the late 19th century**: at one point ~40% of world tin came from Cornwall/Devon 【60†L740-L748】

【62†L1530-L1538】. For example, tin output reached ~10,000 t in 1870–71 【62†L1530-L1538】. In the 20th century, metals declined, and **china clay (kaolin)** dominated. UK china clay production peaked at ~3.3 Mt in 1988 【60†L750-L758】, 90% of which came from Cornwall 【70†L975-L983】. Over 150 Mt of marketable kaolin has been extracted from SW England since the mid-18th century 【70†L975-L983】 (St Austell Granite supplies ~83.5% of this 【70†L975-L983】). Kaolin remains economically vital (~£200 M p.a. 【60†L786-L794】).

Metal mining persists at a smaller scale. In the 1970s–80s, Wheal Jane (copper–tin) boosted tin output to ~5,200 t/yr 【62†L1530-L1538】. However, tin prices collapsed after the 1985 Tin Council crisis, and by 1991 all Cornwall's large tin mines closed except **South Crofty** 【62†L1530-L1538】. Wheal Jane's last full year (1990) yielded 1,041 t Sn in concentrate 【61†L1-L4】. South Crofty (Pool, Camborne) then became the sole major metal producer (~2,000 t/yr) until its closure in 1997 【62†L1542-L1550】. At that time SC held ~3.2 Mt at 1.50% Sn (830 kt ore @1.54% Sn) 【62†L1548-L1554】. Cornwall also yielded ~2 Mt Cu (peak >15,000 t/yr in 1860s) 【62†L1599-L1607】 and significant Pb–Zn, W, Ag as by-products. Many mines (Geevor, East Pool, Caradon, West Devon) remain accessible underground.

Major Deposit Examples

Notable high-value deposit areas include:

- **South Crofty (Pool)**: Historic Sn–Cu mine. *Resource (2026)*: ~8.5 Mt @ ~1.04% Sn, containing ~87.8 kt Sn 【39†L69-L72】. Cornish Metals plans to restart it (targeting ~5,000 tpa tin) 【39†L69-L72】. Infrastr. excellent (road/rail, existing shafts/winzes, processing plant on site). Located in Camborne-Redruth urban area with power and skilled workforce.

- **Redmoor (Callington area):** Tungsten–tin–copper sheeted vein deposit. *Resource (2026):* 17.4 Mt @ 0.65% WO₃ equiv (85.8 kt WO₃, 29.0 kt Sn, 76.3 kt Cu) [64†L59-L60] . Licensed by Cornwall Resources, with scoping study NPV ≈ US\$1.5 B. Good access (near A388/A30, Plymouth port ~25 km) and grid power. Part of the old Callington mining district. Redmoor is likely the **highest-grade undeveloped W deposit in Europe** [81†L80-L83] .
- **Hemerdon (Devon border):** Tungsten–tin open pit (W Tungsten West). *Reserves (historic):* 141.4 kt WO₃ in 113 Mt ore [81†L80-L83] . Recently reopened (Tungsten West), providing local processing. (Hemerdon is just outside Cornwall but its economics and infrastructure benefit Cornwall.)
- **Caradon District (Liskeard):** Polymetallic Cu–Sn–Au veins (Phoenix, South Caradon, Kit Hill, Gunnislake). Historically produced Ag–Pb–Cu–Sn; e.g. 574 t Sn, 268 t WO₃, 852 t Cu at Redmoor/Phoenix (1930s) [64†L105-L107] . Multiple small historic workings. Potential for new targets along granite contact (sheeted veins at Caradon Hill, Kit Hill area) and deep extensions of old lodes.
- **St Austell Granite:** World's premier china clay field (Imerys). Produces ~80% of UK kaolin. Extensive pits and tailings dams exist. Potential new targets: (a) **Lithium in mica** – the granite micas contain up to ~2.5% Li [57†L1883-L1890] , although historically low-grade; (b) **by-product clays/minerals** in deposits and waste dumps [57†L1883-L1890] . Also geothermal potential for Li-rich brines. Site has excellent transport and power (near A390/A30).
- **Kit Hill–East Wheal Rose (Gunnislake):** Historic Cu–Sn–W–As mines. E.g. Wheal Gorland (Skewjack) produced Sn/W/As. Area on granite margin with many veins. Accessibility via A390/A38. Offers potential *brownfield* exploration (old workings, good datasets).
- **Bodmin Moor Granite:** Minor historical clay (Park, Stannon pits) and metal (Trewortha). Possibly hosts deep Sn–Cu structures or Li. Remote terrain (AONB), limited infrastructure, so lower priority.
- **Other Devon–Cornwall Border:** Tamar Valley/Southwest Exeter area near AONBs. Hosts some known lead/zinc (Dolcoath style) and potential for Ni–Cu (Komatiite belt? historically explored). Early-stage interest (e.g. Tamar Valley drilling).

In addition to these, Cornwall has numerous small prospects (Perranporth–St Day quartz tourmaline deposits, Tintagel zone, etc.). Company and government surveys have catalogued hundreds of “MINLOC” occurrences (BGS database).

Spatial Data and Mapping

Extensive digital data exist. BGS 1:50k sheets (341–349) cover Cornwall (lithology, structure). The DETR/ BGS Mineral Resources report (above) compiled these into GIS layers (mineral occurrences, planning consents, constraints) [37†L237-L245] [73†L618-L623] . Modern surveys include the **Tellus South West** project (regional airborne radiometrics/magnetics, stream-soil geochemistry) and G-BASE baseline geochemistry (heavy-mineral concentrates in soils). Cornwall Council provides an online Mineral Safeguarding DPD (maps of safeguarding areas) [76†L81-L90] . **Protected site overlays:** AONBs (Cornwall Coast, Tamar Valley), World Heritage Mining Landscape (UNESCO WHS), Heritage Coast, SAC/ SPA/SSSI (e.g. Lizard complex, Fal/Rame), etc., have been overlaid on all maps [73†L618-L623] . These

constraints (AONB, SSSI, NNR, scheduled monuments etc.) cover significant areas [73†L618-L623] . GIS layers of land cover/land use and transport (roads, rail, grid) are available via Cornwall Council and Ordnance Survey.

For field targeting, high-resolution DEM (LiDAR) exists for much of Cornwall (covering old shafts and pits). Satellite imagery can highlight alteration (kaolin or chlorite alteration zones). Geophysical surveys (ground magnetics, EM) can map concealed veins; gravity surveys can delineate granite boundaries.

Environmental and Land-Use Constraints

Much of Cornwall's uplands and coasts are environmentally sensitive. The **Cornwall & West Devon Mining Landscape** is a UNESCO World Heritage Site, and large areas are AONB or Heritage Coast [73†L618-L623] . National designations (SSSI/SAC) often protect wetlands, heathlands and woodlands within historic mining areas (e.g. Tamar Valley, Carnmenellis Commons). All potential sites must account for ecology (habitats, protected species) and heritage (mine remains, archaeology). For example, the Cornwall AONB covers inland zones (e.g. Bodmin Moor) and South West Coast Path runs along many tin-mining districts. Planning authorities require EcIA and HIA in addition to standard mining permits. Baseline environmental surveys (flora, fauna, water quality) would be integrated into site investigations. Note that BGS mapping already flags many statutory designations (AONB, SSSI, NNR, Scheduled Monuments) in Cornwall [73†L618-L623] .

Infrastructure and Access

Cornwall has reasonable infrastructure near historical mining areas. Major roads (A30/A38/A390) traverse the region. Rail connections exist to Camborne/Redruth and the Looe branch line (though freight use is limited). Deep-sea ports at Falmouth, Plymouth and Hayle – and on the Tamar – can handle minerals. The 275 kV national grid runs across Cornwall, with substations at Camborne, Saltash and Bodmin. Many old mines have onsite facilities: adits/shafts (e.g. South Crofty), processing plants (limited at Wheal Jane), and tailings/waste facilities (e.g. Imerys clay dams). This existing infrastructure can significantly reduce capital cost for new projects. Conversely, some targets (like Bodmin Moor or west Penwith) are more remote and lack roads, which increases exploration cost and community/traffic issues.

Exploration Potential and Critical Minerals

Recent reviews emphasize Cornwall as a **critical minerals province** [39†L21-L27] . Beyond tin/tungsten, attention is on **lithium** and other tech minerals. The granites are rich in Li-bearing micas (muscovite/lepidolite) – St Austell Granite alone contains *hundreds of millions of tonnes* of Li-resource [39†L39-L43] . Two UK companies have declared ~250 Mt combined Li resource in Cornish hard-rock (Cornish Lithium and Cornish Tin & Lithium) [39†L39-L43] . Pilot plants for Li processing (froth flotation of mica, geothermal brine extraction) are already in operation. Other metals (Ni–Co in ultramafic remnants, Sb–As in tin ores) are underexplored [81†L100-L103] . For example, paragenetic gold has been found in Caradon and Gunnislake veins – currently under active drilling by small explorers. **Geothermal energy** (Cornish granite has high heat output) is also being pursued and could provide low-carbon power for future mining.

Given rising metal prices and tech demand, even low-grade resources may become viable. For example, the BGS once judged the St Austell Li “not economically interesting” [57†L1883-L1890] , but today

advances in processing and the UK's strategic needs may change that outlook. **RE-Waste:** Cornwall has millions of tonnes of historic mine waste (tailings, lodes-residuum) that could be reprocessed for critical elements (e.g. Zn skarns, As residues, Co in blackwall), following circular-economy principles [81†L100-L103] .

Target Ranking (Table)

The table below ranks the top exploration targets in Cornwall by estimated value and feasibility. Criteria include known resource size, commodity criticality, infrastructure, and risk factors (e.g. environmental constraints). Ranks are indicative and based on a combination of resource tonnage/grade and strategic importance.

Rank	Target Area	Commodities	Key Attributes (Reserves, Access, Constraints)	Justification
1	Redmoor (Callington)	W, Sn, Cu (Ag, Pb)	<i>Resource (2026):</i> 17.4 Mt @ 0.65% WO ₃ eq (85.8 kt WO ₃ , 29 kt Sn, 76 kt Cu) [64†L59-L60] . Well-studied (JORC) deposit. Near A38/A30, Plymouth port. Low AONB impact.	Highest-grade undeveloped tungsten deposit in Europe [81†L80-L83] ; co-products (Sn, Cu) add value. Strong economic case (NPV~\$1.5 B).
2	South Crofty (Pool)	Sn (Cu, Ag, Zn)	<i>Resource:</i> ~8.5 Mt @ ~1.0% Sn (87.8 kt Sn) [39†L69-L72] . Existing mine infrastructure, grid power, skilled labor. Located in Camborne-Redruth (former industrial area). Moderate AONB issues (none).	Contains ~6 Mt ore with 87.8 kt tin [39†L69-L72] . Reopening would secure UK tin supply and leverage existing shafts/plant. Low environmental sensitivity.
3	St Austell Granite	Kaolin, Li	<i>Production:</i> Largest global kaolin mine (Imerys). <i>Potential:</i> Li in mica (2.5% in places [57†L1883-L1890]), by-product feldspar. Excellent roads (A390/A30) & port nearby. High density of protected sites (NSAs, SSSIs).	150 Mt kaolin extracted [70†L975-L983] ; kaolin of international importance [76†L81-L90] . Emerging lithium projects indicate large Li resource [39†L39-L43] .

Rank	Target Area	Commodities	Key Attributes (Reserves, Access, Constraints)	Justification
4	Tungsten West (Devon border)	W, Sn	Hemerdon tungsten mine (Tungsten West): reserves ~141 kt WO ₃ 【81†L80-L83】. Full infrastructure (road, rail, processing plant). In nearby Devon, but benefits Cornish supply chains. Low new exploration needed (brownfields).	UK's 2nd-largest W deposit 【81†L80-L83】. Recommissioned plant gives near-term production (planning consents in place). Critical W source (5% global).
5	Caradon District	Sn, Cu (Au, Ag)	Historic mines (South Caradon, Kit Hill, Phoenix). Known lodes – ~574 t Sn, 852 t Cu produced historically 【64†L105-L107】. Good access (A38). Proximity to granite contact. Moderate environmental oversight (Area of Great Landscape Value).	Polymetallic veins with tin/copper: potential undiscovered extensions. Gold/silver occurrences add upside. Historically productive; follow-up of past mines is cost-effective.
6	Kit Hill–Gunnislake	Cu, Sn, W, As (Pb, Ag)	Old mines (South Crofty/Gunnislake field). <i>Past</i> : Cu–Sn–W/As on arsenopyrite veins. Holds Redmoor project. A390 corridor, close to Saltash. Moderate : partly within AONB (Cornwall Coast).	Continuity with Redmoor resources. Known high-grade pods historically. Viable for continued exploration (e.g. Johnson's lode, Phoenix lode).
7	Bodmin Moor Granite	Sn, Cu, Li (China clay)	<i>Past</i> : small tin/copper mines (e.g. Cadson). <i>China clay</i> : Park/Stannon pits (operation). Large blocks of unmined granite. High : Moors are designated (AONB, SACs). Remote with little infrastructure.	Limited historic production. Potential for concealed LCT pegmatites (Li) along De Lank, or deeper vein extensions. However, environmental and access challenges are high.

Rank	Target Area	Commodities	Key Attributes (Reserves, Access, Constraints)	Justification
–	Other prospects	Ni, Co, Sb, Au, Zn	Various underexplored (e.g. serpentinite bodies, granitoid margins). Local small occurrences (e.g. Seaton tin, St. Dennis antimony). Typically low-signal, requiring advanced targeting.	Given low grades, ranked lower but noted for future opportunity if tech or markets change (e.g. Ni/Co in podiform chromite, Zn at Wheal Jane tailings).

Notes: Li in mica and by-products may shift priorities as critical minerals demand grows [39†L39-L43] [57†L1883-L1890] . Constraints: many targets lie near natural reserves or within the Cornwall & West Devon WHS. Stakeholder acceptability and permitting effort differ by site.

Proposed Methods

Literature and Data Review: Compile historical mining records (e.g. BGS Memoirs [33†L153-L161] , company reports, Cornish Mining archives). Digitise old mine plans. Gather existing geoscience datasets (BGS 1:50k maps, Tellus SW geochemistry, LiDAR topography, BGS MINLOC database). Use GIS to overlay geology, mineral occurrences, geochem anomalies, tenure and constraints.

Field Reconnaissance: Geological mapping of vein structures, sampling of outcrops, mine spoil, and mill tailings for geochemical assays (ICP-MS for trace metals, including Li, Sn, Cu, W, As, Sb, REE). Soil and stream sediment sampling over target districts, with dense grids (~200 m) to refine mineralization footprints. Heavy-mineral till sampling at granitic contacts to trace tin-tungsten halos (cassiterite indicators).

Geophysical Surveys:

- *Airborne:* High-resolution magnetics/radiometrics to map granite edges and alteration. Heliborne EM to detect conductive sulphide bodies in target areas.
- *Ground:* IP/resistivity surveys across known veins to image sulphide zones (especially for tin/copper targets like South Crofty extensions). Gravity surveys across fault zones to delineate buried veins (e.g. Caradon). For geothermal prospects, magnetotellurics can image deep heat sources.
- *Downhole:* If drill holes are sunk, perform borehole electromagnetics (BHEM) to locate extensions of ore below holes.

Drilling and Sampling: On priority targets (Rank 1–3), plan core drilling to test resource extensions. For Redmoor and South Crofty, infill drilling in resource models; for St Austell Li, target fresh granite with high-Mg mica zones. Drill core to be logged geotechnically and sampled for multi-element assays (Au/Ag by fire assay + ICP). Dating (e.g. U-Pb on cassiterite, Ar-Ar on mica) can refine ore formation models. Analytical methods include XRD (for mineral identification), ICP-MS/OES (full element suite), and possibly QEMSCAN automated mineralogy for complex sulphides.

Laboratory: Samples analyzed for major, trace, rare-earth elements. Use high-sensitivity ICP-MS for Li and REE in micas and brines. Ore microscopy and electron microprobe to characterize ore minerals (e.g.

wolframite, cassiterite, arsenopyrite). Geometallurgical tests (crushing/gravity separation, flotation) on representative samples to guide future feasibility.

Data Integration: Use 3D geological modelling (Leapfrog, GEMS) to combine geology, geophysics and drilling into target models. Mineral resource estimation software (MineSight/Petrex) for inferred resource blocks. All spatial data in GIS to maintain line-of-sight to constraints (SACs, etc.) and infrastructure.

Permitting and Stakeholder Engagement

This project will require a proactive permitting plan. Mineral extraction in England is regulated by the Planning Acts and Environmental Permitting. Major steps include: obtaining mining planning permission from Cornwall Council (with an Environmental Impact Assessment for any new or enlarged operations), and Environment Agency permits for water discharge and waste (if needed). For below-ground works, a Mining Waste Directive permit (or UK equivalent) applies. Protected heritage sites (scheduled mines) require Archaeological consent. Early liaison with Natural England (for SSSI/SAC), the AONB boards, and Historic England is essential to identify issues. Public consultations will be held in local communities (e.g. Camborne, Callington, St Austell towns) and with stakeholder groups (farmers, fishermen, tourism operators, slate workers' unions, etc.). Engagement with Camborne School of Mines (for technical advice) and the Cornwall Mining & Geo-Resources Alliance (CMGA) can facilitate local acceptance. Work plans will incorporate ecological and archaeological baseline surveys as part of consent applications.

Risk Assessment

Key risks include: **Geological risk** (target may not host mineable ore), **commodity price risk** (tin/tungsten prices fluctuate), **permitting risk** (delay or refusal due to environment/heritage), and **community opposition** (NIMBY concerns in scenic areas). To mitigate, the plan includes early-stage wide screening (desk and surface surveys) to derisk before expensive drilling. Adaptive management: e.g. if invasive field works are required in sensitive areas, we will use small-window diamond drills and reinstatement. Financial risk is managed by staged funding (milestones). A monitoring plan (water quality, dust, noise) will be prepared to comply with regulations and reassure local residents. Emergency response plans for accidents (flooding, contaminant spills) will be drafted.

Budget and Timeline

A provisional budget (ballpark) is given in low/medium/high categories:

- **Low (~£0.5–1 M):** Desktop review, sample assays, limited fieldwork, and preliminary geophysics over one or two areas.
- **Medium (~£2–5 M):** Adds multiple drill holes (5–10 holes), detailed mapping, full geochem grids, and IP surveys at top targets.
- **High (~£5–10 M+):** Extensive multi-season drilling campaigns (20+ holes), comprehensive geophysical campaigns (airborne EM plus ground surveys), and advanced laboratory testing.

Timeline (5-year): Year 1 – Data compilation, initial field sampling, stakeholder outreach. Year 2 – Detailed geophysics and geochemistry (Tellus follow-up). Year 3 – Drill permitting and drilling at Rank 1 targets (Redmoor, South Crofty). Year 4 – Additional drilling (Rank 2–3 targets), lab analysis, resource

modeling. Year 5 – Final evaluation, permitting applications for development (if warranted), and reporting. Milestones include resource update announcements (e.g. as with Redmoor’s 2026 MRE [64†L59-L60]) and permitting approvals. No absolute deadline is set; schedule is driven by funding and data results.

Project Workflow (Mermaid)

flowchart LR

A[Data Compilation & Review] --> B[Stakeholder Engagement & Permitting Prep]

B --> C[Field Reconnaissance (mapping, sampling)]

C --> D[Geophysical Surveys (airborne, ground)]

D --> E[Target Refinement (modeling)]

E --> F[Drilling & Lab Analysis]

F --> G[Resource Estimation & Economic Analysis]

G --> H[Environmental & Safety Planning]

H --> I[Reporting & Proposal Draft]

I --> J{Next Steps}

J --> |Resource positive| K[Development Plan]

J --> |Negative/Flat| L[Abort or Pivot to New Targets]

References

Key data and maps are drawn from BGS publications and datasets (e.g. BGS Mineral Reconnaissance programme reports [73†L618-L623] [62†L1530-L1538]), Cornwall Council planning documents [76†L81-L90] , and recent industry reports and peer sources [39†L21-L27] [64†L59-L60] . These include production figures, resource estimates, and geological descriptions as cited above.
